Portfolio 2: Query Processing and Optimisation

These queries were used to bring out useful University records.

```
USE UniversityManagementDB;

-- Query 1: Retrieve all students
SELECT * FROM Student;


-- Query 2: Retrieve students and their departments
SELECT
   s.matric_number,
   s.full_name,
   d.department_name
FROM Student s
JOIN Department d
   ON s.department_id = d.department_id;


-- Query 3: Retrieve students who scored above 80
SELECT
   s.full_name,
   c.course_code,
   e.total_score
FROM Enrollment e
JOIN Student s ON e.student_id = s.student_id
JOIN CourseOffering co ON e.offering_id = co.offering_id
JOIN Course c ON co.course_id = c.course_id
WHERE e.total_score > 80;
```

This query uses EXPLAIN to tell MySQL how it plans to execute a query.

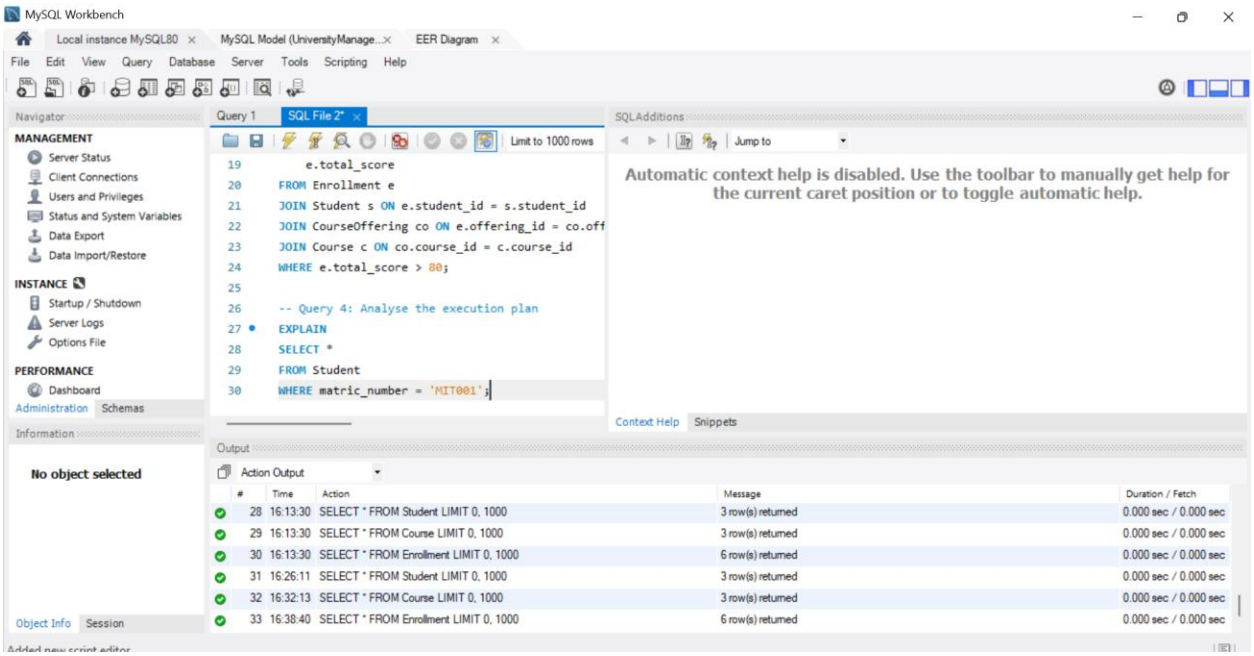
```
-- Query 4: Analyse the execution plan

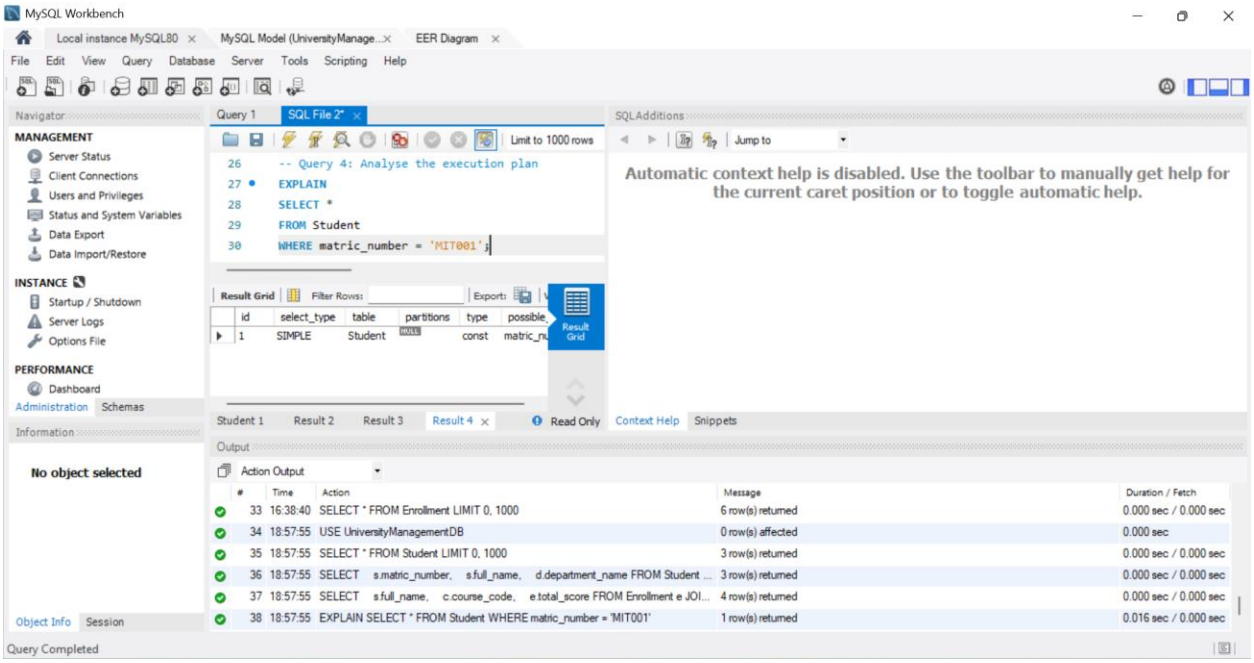
EXPLAIN

SELECT *

FROM Student

WHERE matric_number = 'MIT001';
```





Demonstration of indexing and Query Optimization

A comparison to know how MySQL executes a query before creating an Index and after creating an Index.

USE UniversityManagementDB;

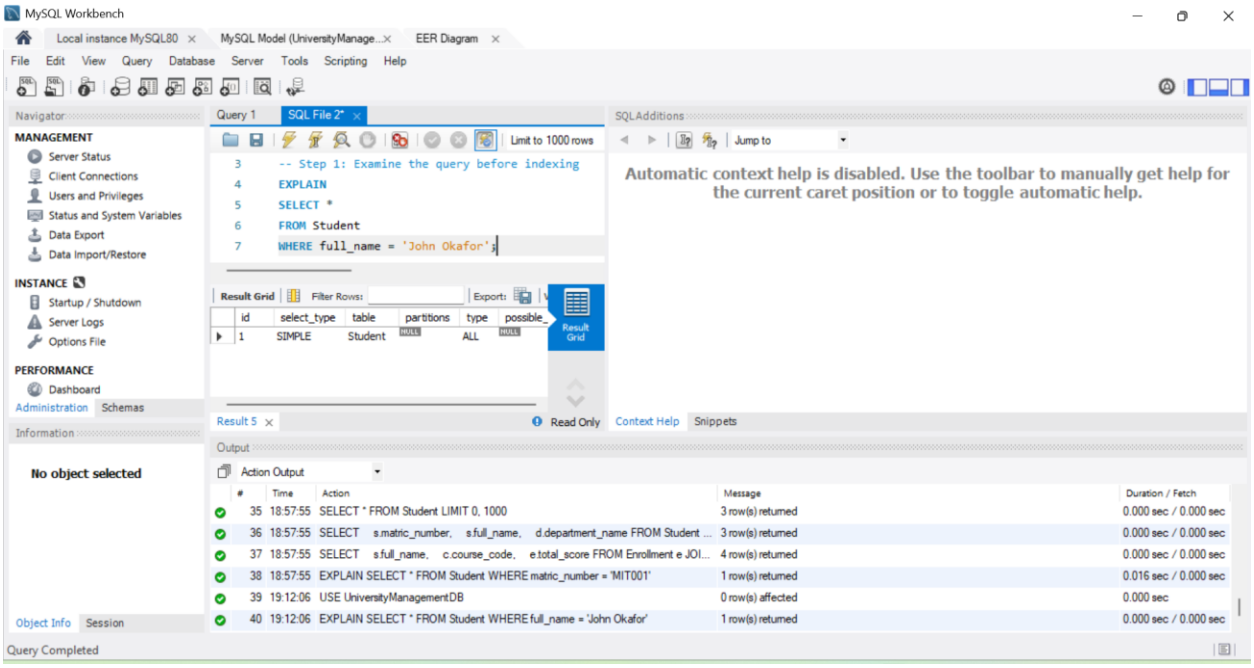
-- Step 1: Examine the query before indexing

EXPLAIN

SELECT *

FROM Student

WHERE full_name = 'John Okafor';

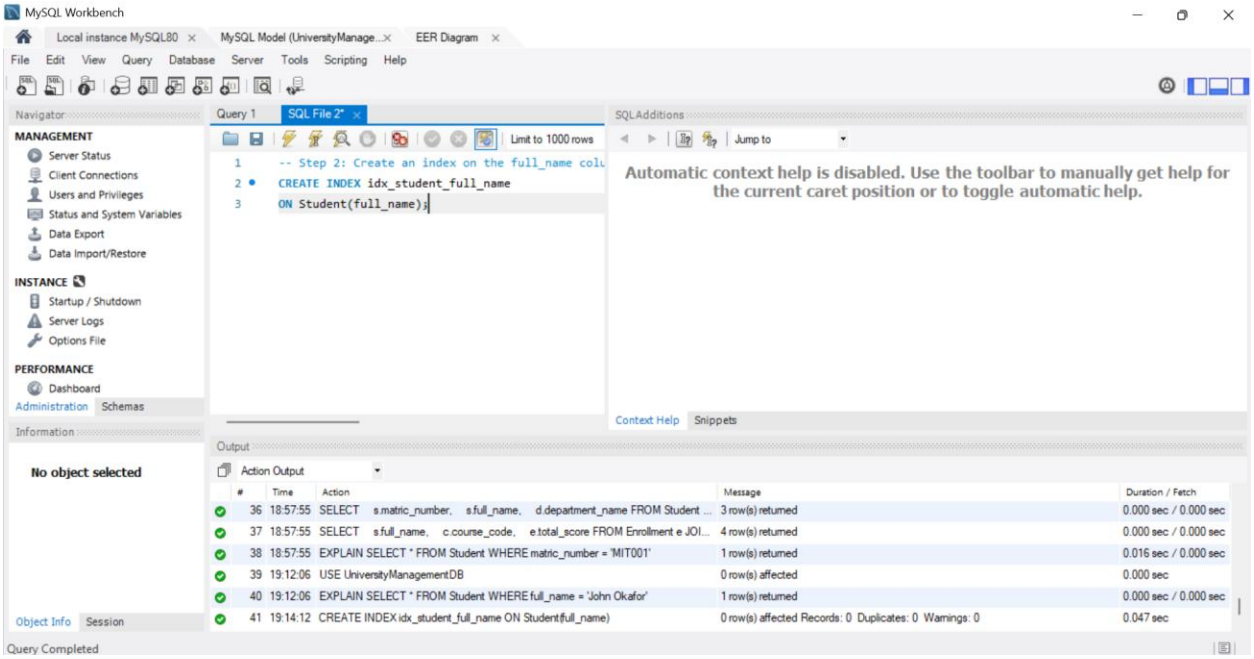


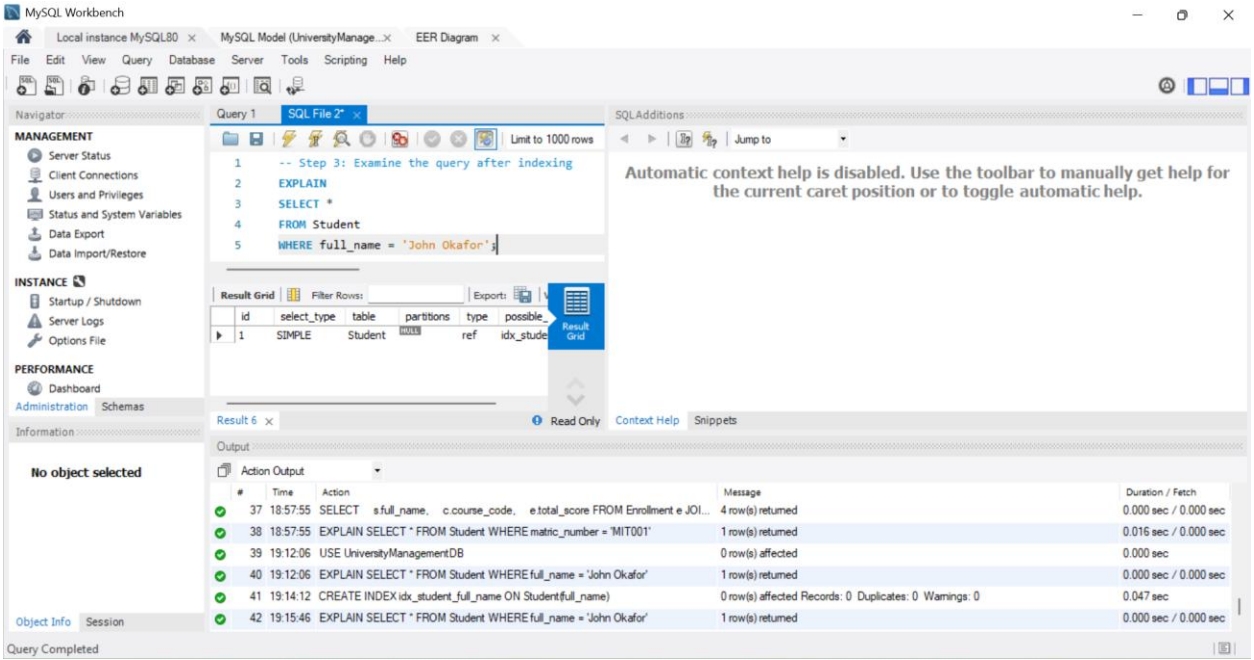
Creating an Index

-- Step 2: Create an index on the full_name column

CREATE INDEX idx_student_full_name

ON Student(full_name);


MySQL Workbench
Local instance MySQL80
MySQL Model (UniversityManage...
EER Diagram
File Edit View Query Database Server Tools Scripting Help
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INSTANCE
Startup / Shutdown
Server Logs
Options File
PERFORMANCE
Dashboard
Administration Schemas
Information
No object selected
Object Info Session
Query 1
SQL File 2*
Limit to 1000 rows
1 -- Step 2: Create an index on the full_name colu
2 CREATE INDEX idx_student_full_name
3 ON Student(full_name);
SQLAdditions
Jump to
Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.
Context Help Snippets
Output
Action Output
Time Action Message Duration / Fetch
36 18:57:55 SELECT s.matric_number, s.full_name, d.department_name FROM Student ... 3 row(s) returned 0.000 sec / 0.000 sec
37 18:57:55 SELECT s.full_name, c.course_code, e.total_score FROM Enrollment e JOI... 4 row(s) returned 0.000 sec / 0.000 sec
38 18:57:55 EXPLAIN SELECT * FROM Student WHERE matric_number = 'MIT001' 1 row(s) returned 0.016 sec / 0.000 sec
39 19:12:06 USE UniversityManagementDB 0 row(s) affected 0.000 sec
40 19:12:06 EXPLAIN SELECT * FROM Student WHERE full_name = 'John Okafor' 1 row(s) returned 0.000 sec / 0.000 sec
41 19:14:12 CREATE INDEX idx_student_full_name ON Student(full_name) 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0 0.047 sec
Query Completed


MySQL Workbench
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Dashboard
Administration Schemas
Information
No object selected
Object Info Session
Query 1
SQL File 2*
Limit to 1000 rows
1 -- Step 3: Examine the query after indexing
2 EXPLAIN
3 SELECT *
4 FROM Student
5 WHERE full_name = 'John Okafor';
Result Grid Filter Rows: Export:
id select_type table partitions type possible_
1 SIMPLE Student NULL ref idx_stude
Result Grid
Result 6
Read Only
SQLAdditions
Jump to
Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.
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41 19:14:12 CREATE INDEX idx_student_full_name ON Student(full_name) 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0 0.047 sec
42 19:15:46 EXPLAIN SELECT * FROM Student WHERE full_name = 'John Okafor' 1 row(s) returned 0.000 sec / 0.000 sec
Query Completed

Creating a Composite Index

```
USE UniversityManagementDB;

CREATE INDEX idx_student_department_level
ON Student(department_id, study_level);

EXPLAIN
SELECT *
FROM Student
WHERE department_id = 1
AND study_level = 800;
```

